

Two Knights Problem Explanation

Today 10:07 AM

Absolutely. This is a **combinatorics + counting** problem. The key is to understand:

How many ways can we choose **2 squares** on a $k \times k$ chessboard such that the two knights **do not attack each other**?

Let's build it from the basics.

1. What is the question asking?

For every:

$$k = 1, 2, 3, \dots, n$$

we have a $k \times k$ chessboard.

We need to place **exactly two knights** on the board such that they **cannot attack each other**.

For example, when $k = 2$:

```
+---+---+
|   |   |
+---+---+
|   |   |
+---+---+
```

There are 4 squares.

We need to choose 2 of them.

Number of ways to choose 2 squares:

$$\binom{4}{2} = 6$$

Can two knights attack each other on a 2×2 board?

No.

A knight moves:

```
2 squares in one direction
+
1 square perpendicular
```

So it needs at least a 2×3 or 3×2 area to attack.

Therefore all 6 arrangements are valid.

Hence:

$$answer(2) = 6$$

2. First count: total ways to put two knights

A $k \times k$ board has:

$$k^2$$

squares.

We need to choose **2 different squares**.

Therefore:

$$\text{Total ways} = \binom{k^2}{2}$$

Using the combination formula:

$$\binom{m}{2} = \frac{m(m-1)}{2}$$

we get:

$$\boxed{\text{Total} = \frac{k^2(k^2 - 1)}{2}}$$

But this includes arrangements where the knights **attack each other**.

So we need to subtract those.

3. When can two knights attack?

A knight moves like this:

```

  K
  |
K---+
```

More accurately, from one square it moves:

- 2 rows + 1 column
- OR
- 1 row + 2 columns

So the attacking pattern occupies either:

2 × 3

or

3 × 2

For example:

```

K . .
. . K
```

These two positions are a knight's move apart.

4. Count attacking pairs

Let's first look at a 2 × 3 rectangle.

```

+---+---+---+
|   |   |   |
+---+---+---+
|   |   |   |
+---+---+---+
```

How many attacking knight pairs are inside this rectangle?

There are **2**.

They look like:

```

K . .
. . K
```

and

```

. . K
K . .
```

So every 2 × 3 rectangle gives:

2

attacking pairs.

Similarly, every 3 × 2 rectangle gives:

2

attacking pairs.

5. How many 2 × 3 rectangles exist?

On a k × k board:

For a 2 × 3 rectangle:

- vertically, it can start in $k - 1$ positions
- horizontally, it can start in $k - 2$ positions

Therefore:

$$(k - 1)(k - 2)$$

such rectangles exist.

Each has 2 attacking pairs.

Therefore:

$$2(k - 1)(k - 2)$$

attacking pairs.

6. What about 3×2 rectangles?

Same logic:

$$(k - 2)(k - 1)$$

rectangles.

Each has 2 attacking pairs.

Therefore:

$$2(k - 1)(k - 2)$$

again.

So total attacking pairs:

$$2(k - 1)(k - 2) + 2(k - 1)(k - 2)$$

which is:

$$\boxed{4(k - 1)(k - 2)}$$

7. Therefore the answer

We had:

$$\text{Total pairs} = \frac{k^2(k^2 - 1)}{2}$$

and:

$$\text{Attacking pairs} = 4(k - 1)(k - 2)$$

Therefore:

$$\boxed{\text{answer}(k) = \frac{k^2(k^2 - 1)}{2} - 4(k - 1)(k - 2)}$$

That's the entire mathematical solution.

8. Let's verify the example

$k = 1$

Only one square.

Cannot put two knights.

$$\frac{1^2(1^2 - 1)}{2} = 0$$

Answer:

$$\boxed{0}$$

k = 2

Total:

$$\frac{2^2(2^2 - 1)}{2}$$

$$= \frac{4 \times 3}{2} = 6$$

Attacking:

$$4(2 - 1)(2 - 2) = 0$$

Therefore:

$$6 - 0 = \boxed{6}$$

k = 3

Total:

$$\frac{9(8)}{2} = 36$$

Attacking:

$$4(2)(1) = 8$$

Therefore:

$$36 - 8 = \boxed{28}$$

Matches the example.

k = 4

Total:

$$\frac{16(15)}{2} = 120$$

Attacking:

$$4(3)(2) = 24$$

Therefore:

$$120 - 24 = \boxed{96}$$

Matches.

9. Why don't we care about the order of the knights?

This is important because you were asking about combinations in your previous question.

Suppose we choose squares A and B.

Knight 1 → A
Knight 2 → B

and:

Knight 1 → B
Knight 2 → A

These represent the **same placement**.

The knights are identical.

So we use:

$$\binom{k^2}{2}$$

rather than:

$$k^2(k^2 - 1)$$

because **order doesn't matter**.

10. Another way to understand the attacking formula

This is probably the most important part to visualize.

A knight attack always fits inside one of these:

2 × 3

or

3 × 2

For a $k \times k$ board:

2 × 3

Number of placements:

$$(k - 1)(k - 2)$$

Each rectangle has 2 attacking pairs:

$$2(k - 1)(k - 2)$$

3 × 2

Again:

$$2(k - 1)(k - 2)$$

Together:

$$4(k - 1)(k - 2)$$

That's where the mysterious 4 comes from.

11. Code

Since $n \leq 10000$, we can simply calculate every k from 1 to n .

```
C++
#include <bits/stdc++.h>
using namespace std;

int main() {
    int n;
    cin >> n;

    for (long long k = 1; k <= n; k++) {
        long long total = k * k * (k * k - 1) / 2;
        long long attacking = 4 * (k - 1) * (k - 2);
        cout << total - attacking << '\n';
    }

    return 0;
}
```

Complexity

We calculate one formula for each k .

$$O(n)$$

Space:

$$O(1)$$

The formula to remember

For a $k \times k$ board:

$$\text{Answer} = \binom{k^2}{2} - 4(k-1)(k-2)$$

Think of it as:

All ways to choose 2 squares – ways where knights attack each other.

That is the core idea of **CSES Two Knights**.

Yes. Let's **ignore the knight attacking condition for now** and understand only:

How many ways can we place 2 knights on a $k \times k$ board?

This is simply a **choose 2 squares from all available squares** problem.

1. 1×1 board

There is only **1 square**:

```
+---+
| K |
+---+
```

We need to place 2 knights, but there is only 1 square.

So:

0

2. 2×2 board

There are:

$$2 \times 2 = 4$$

squares.

Let's number them:

```
+---+---+
| 1 | 2 |
+---+---+
| 3 | 4 |
+---+---+
```

We need to select **2 squares**.

Possible placements:

```
1,2
1,3
1,4
2,3
2,4
3,4
```

So there are:

6

ways.

Notice that:

```
1,2
```

and

```
2,1
```

are **the same placement**.

We don't care which knight is called "knight 1" or "knight 2".

Therefore we use combinations:

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$$\begin{aligned} & \binom{4}{2} \\ &= \frac{4 \times 3}{2} \\ &= \boxed{6} \end{aligned}$$

3. 3 × 3 board

There are:

$$3 \times 3 = 9$$

squares.

Number them:

```
+---+---+---+
| 1 | 2 | 3 |
+---+---+---+
| 4 | 5 | 6 |
+---+---+---+
| 7 | 8 | 9 |
+---+---+---+
```

We choose any 2 of these 9 squares.

Therefore:

$$\begin{aligned} & \binom{9}{2} \\ &= \frac{9 \times 8}{2} \\ &= \boxed{36} \end{aligned}$$

So there are **36 total ways** to place two knights.

⚠ This is **before removing attacking positions**.

4. 4 × 4 board

There are:

$$4 \times 4 = 16$$

squares.

We choose any 2:

$$\begin{aligned} & \binom{16}{2} \\ &= \frac{16 \times 15}{2} \\ &= \boxed{120} \end{aligned}$$

So there are **120 total placements**.

5. Why exactly $nC2$?

Suppose we have N squares.

For the **first knight**, we have:

$$N$$

choices.

For the **second knight**, we have:

$$N - 1$$

choices.

So:

$$N(N - 1)$$

But this counts every placement **twice**.

For example:

Knight A → square 1
Knight B → square 2

and:

Knight A → square 2
Knight B → square 1

are physically the same board.

Therefore divide by 2:

$$\frac{N(N - 1)}{2}$$

Since a $k \times k$ board has:

$$N = k^2$$

we get:

$$\frac{k^2(k^2 - 1)}{2}$$

6. Small examples together

Board	Squares	Total ways to place 2 knights
1 × 1	1	0
2 × 2	4	6
3 × 3	9	36
4 × 4	16	120
5 × 5	25	300

For example, for 5 × 5:

$$\frac{25 \times 24}{2} = 300$$

So there are **300 ways to put two knights anywhere**.

7. Very important distinction

For CSES Two Knights, there are actually **two steps**:

Step 1 — Put two knights anywhere

$$\frac{k^2(k^2 - 1)}{2}$$

Step 2 — Remove the positions where they attack

$$4(k - 1)(k - 2)$$

Therefore:

$$\text{valid ways} = \frac{k^2(k^2 - 1)}{2} - 4(k - 1)(k - 2)$$

For example, 3 × 3:

Total placements = 36
Attacking placements = 8
Valid = 36 - 8

= 28

That's why the CSES answer for $k = 3$ is **28**, not 36.

Exactly — this is the **key step** for understanding why the attacking count is $4(n-1)(n-2)$.

Let's focus only on:

How many 2×3 rectangles can fit inside an $n \times n$ matrix?

1. Start with a small example: 4×4

Imagine:

	C1	C2	C3	C4
R1	☐	☐	☐	☐
R2	☐	☐	☐	☐
R3	☐	☐	☐	☐
R4	☐	☐	☐	☐

We want to place a 2×3 rectangle:

☐	☐	☐
☐	☐	☐

2. Count the row positions

Our rectangle needs **2 rows**.

The 4×4 board has 4 rows.

Where can the 2-row rectangle start?

Start at row 1

R1	☒	☒	☒
R2	☒	☒	☒
R3	☐	☐	☐
R4	☐	☐	☐

Start at row 2

R1	☐	☐	☐
R2	☒	☒	☒
R3	☒	☒	☒
R4	☐	☐	☐

Start at row 3

R1	☐	☐	☐
R2	☐	☐	☐
R3	☒	☒	☒
R4	☒	☒	☒

It **cannot** start at row 4 because we need 2 rows.

So:

$$4 - 2 + 1 = 3$$

possible row positions.

Or simply:

$$n - 1$$

3. Count the column positions

Our rectangle needs **3 columns**.

The 4×4 board has 4 columns.

Where can it start?

Start at column 1

☒	☒	☒	☐
☒	☒	☒	☐

Start at column 2



It cannot start at column 3 because we need 3 columns.

So:

$$4 - 3 + 1 = 2$$

possible column positions.

Or:

$$n - 2$$

4. Combine row and column choices

For every possible row position, we can choose every possible column position.

So:

Number of 2×3 rectangles

is:

$$(n - 1)(n - 2)$$

For 4×4 :

$$\begin{aligned} (4 - 1)(4 - 2) \\ = 3 \times 2 \\ = 6 \end{aligned}$$

So there are **6 different 2×3 rectangles** inside a 4×4 board.

5. Let's actually see the 6

Board:

```
1 2 3 4
5 6 7 8
9 10 11 12
13 14 15 16
```

Position 1

```
1 2 3 4
5 6 7 8
9 10 11 12
13 14 15 16
```

Rectangle:

```
1 2 3
5 6 7
```

Position 2

```
1 2 3 4
5 6 7 8
9 10 11 12
13 14 15 16
```

Rectangle:

```
2 3 4
6 7 8
```

Those are the two positions for the **first two rows**.

Now move the rectangle down one row.

Position 3

```
5 6 7
9 10 11
```

Position 4

```
6 7 8
10 11 12
```

And move down again.

Position 5

```
9 10 11
13 14 15
```

Position 6

```
10 11 12
14 15 16
```

So:

6

6. General $n \times n$

Now forget 4.

For an $n \times n$ board:

Rectangle needs 2 rows

Number of possible positions:

$$n - 2 + 1 = n - 1$$

Rectangle needs 3 columns

Number of possible positions:

$$n - 3 + 1 = n - 2$$

Therefore:

$(n - 1)(n - 2)$

different 2×3 rectangles.

7. Why do we also need 3×2 ?

Because a knight can attack in **both orientations**.

A knight attack looks like:

```
K . .
. . K
```

This fits inside:

```
2 x 3
```

But it can also be rotated:

```
K .
. .
. K
```

which fits inside:

```
3 x 2
```

Therefore we count both:

$$2 \times 3$$

and:

$$3 \times 2$$

Each has:

$$(n-1)(n-2)$$

positions.

So total rectangles:

$$2(n-1)(n-2)$$

8. But why does the final formula have 4?

Because **each** 2×3 rectangle contains **2 attacking knight pairs**.

For example:

```
K . .
. . K
```

and:

```
. . K
K . .
```

So:

$$\text{attacking pairs} = 2 \times \underbrace{(n-1)(n-2)}_{2 \times 3} + 2 \times \underbrace{(n-1)(n-2)}_{3 \times 2}$$

Therefore:

$$= 4(n-1)(n-2)$$

The whole idea in one line:

$$\underbrace{(n-1)(n-2)}_{2 \times 3 \text{ positions}} \times \underbrace{2}_{\text{attacks per rectangle}} \times \underbrace{2}_{2 \times 3 \text{ and } 3 \times 2} = 4(n-1)(n-2)$$

That's exactly where the $4(n-1)(n-2)$ in Two Knights comes from.

Exactly. Now you can explain the whole problem in a very natural way. Think of it as **two separate questions**.

1. First: How many ways can I place 2 knights anywhere?

For an $n \times n$ board, there are:

$$n^2$$

squares.

We need to choose **any 2 squares** for the two knights.

So:

$$\begin{aligned} \text{Total ways} &= \binom{n^2}{2} \\ &= \frac{n^2(n^2-1)}{2} \end{aligned}$$

Why divide by 2?

Because the knights are identical. Choosing:

```
Knight A → square 1
Knight B → square 2
```

is the same board as:

```
Knight A → square 2
Knight B → square 1
```

So we count each placement only once.

2. Then: Which of these placements are invalid?

The problem says the knights **must not attack each other**.

A knight attacks another knight when their positions have a difference of:

- 2 rows and 1 column, or
- 1 row and 2 columns.

This means every attacking pair fits inside either:

2×3

or:

3×2

3. Count the 2×3 positions

In an $n \times n$ board:

- 2 rows can be placed in $n - 1$ ways.
- 3 columns can be placed in $n - 2$ ways.

Therefore:

$$(n - 1)(n - 2)$$

different 2×3 blocks.

Each 2×3 block contains **2 attacking pairs**.

So:

$$2(n - 1)(n - 2)$$

attacking pairs from 2×3 .

The same thing happens for 3×2 :

$$2(n - 1)(n - 2)$$

Therefore total attacking pairs:

$$4(n - 1)(n - 2)$$

4. Finally, subtract invalid from total

So our thinking is simply:

$$\text{Valid placements} = \text{All placements} - \text{Attacking placements}$$

Therefore:

$$\frac{n^2(n^2 - 1)}{2} - 4(n - 1)(n - 2)$$

In words, you can explain it like this:

First, I calculate all possible ways to place two knights on the board, without caring whether they attack each other. Since there are n^2 squares, I choose any two of them, giving $n^2 C 2$ possibilities.

Then I remove the invalid placements where the knights attack each other. A knight attack always fits inside a 2×3 or 3×2 rectangle. There are $(n - 1)(n - 2)$ positions for each orientation, and each rectangle contains 2 attacking pairs. Therefore there are $4(n - 1)(n - 2)$ attacking placements.

So the final answer is all possible placements minus all attacking placements.

That is a **very good interview/DSA explanation** because you're explaining the *reasoning* instead of just throwing out the formula.

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